

Basic Networking Skills for Digital Forensics

Idriss Muhammad Abdullahi

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Fellowship in Web Application Security & Digital Forensics
Computer Forensics Case Studies

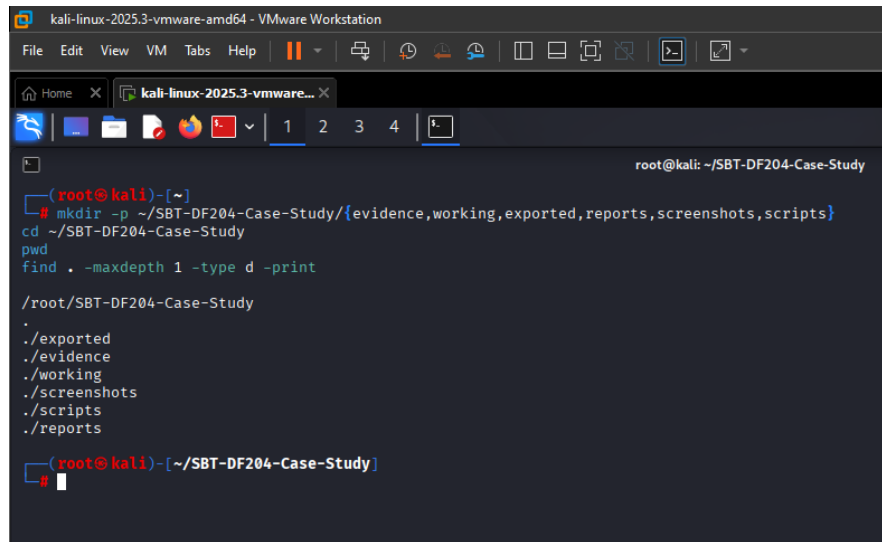
04/10/2026

1. Introduction

This report examines evidence relating to a series of harassing emails sent to Chemistry teacher Lily Tuckrige. The objective is to determine whether network, email, and packet-capture evidence support attribution of the activity to a student enrolled in Chemistry 109 while maintaining forensic integrity and recognising evidential limitations.

2. Evidence Preservation

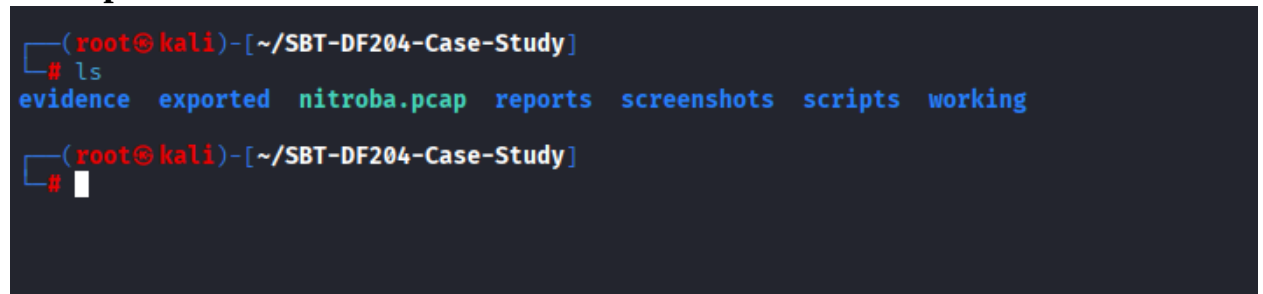
2.1 Folder Structure Creation



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF204-Case-Study
(root@kali)~[~]
# mkdir -p ~/SBT-DF204-Case-Study/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF204-Case-Study
pwd
find . -maxdepth 1 -type d -print

/root/SBT-DF204-Case-Study
.
./exported
./evidence
./working
./screenshots
./scripts
./reports
(root@kali)~[~/SBT-DF204-Case-Study]
```

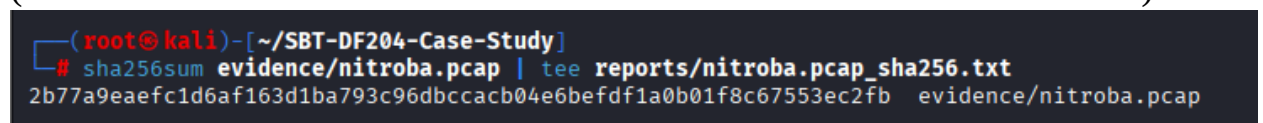
2.2 Captured file downloaded



```
(root@kali)~[~/SBT-DF204-Case-Study]
# ls
evidence  exported  nitroba.pcap  reports  screenshots  scripts  working
(root@kali)~[~/SBT-DF204-Case-Study]
```

2.3 Hashing (Sha256sum)

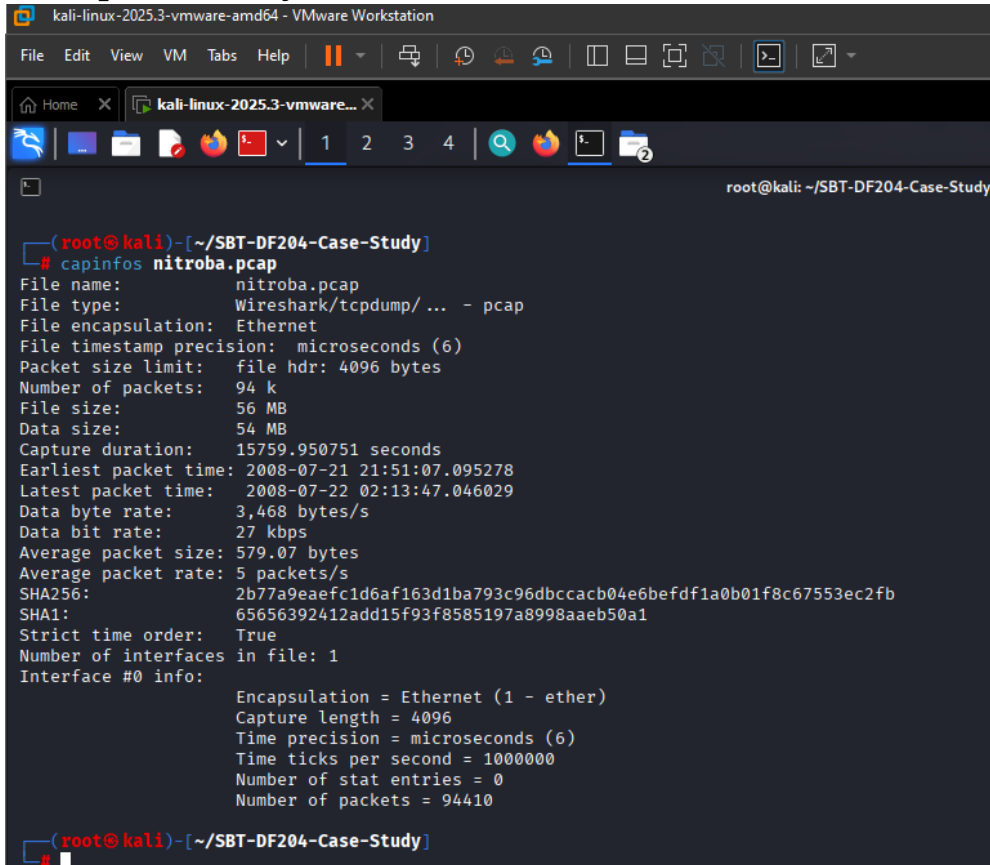
Original Captured packet obtained, a copy created and preserved for analysis (2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb)



```
(root@kali)~[~/SBT-DF204-Case-Study]
# sha256sum evidence/nitroba.pcap | tee reports/nitroba.pcap_sha256.txt
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb evidence/nitroba.pcap
```

Hashes matches and all analysis will be on the working file (No modification will be made for the original file)

2.4 Capinfos Analysis



```
(root@kali)-[~/SBT-DF204-Case-Study]
# capinfos nitroba.pcap
File name: nitroba.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 4096 bytes
Number of packets: 94 k
File size: 56 MB
Data size: 54 MB
Capture duration: 15759.950751 seconds
Earliest packet time: 2008-07-21 21:51:07.095278
Latest packet time: 2008-07-22 02:13:47.046029
Data byte rate: 3,468 bytes/s
Data bit rate: 27 kbps
Average packet size: 579.07 bytes
Average packet rate: 5 packets/s
SHA256: 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
SHA1: 65656392412add15f93f8585197a8998aaeb50a1
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
  Encapsulation = Ethernet (1 - ether)
  Capture length = 4096
  Time precision = microseconds (6)
  Time ticks per second = 1000000
  Number of stat entries = 0
  Number of packets = 94410

(root@kali)-[~/SBT-DF204-Case-Study]
#
```

The pcap file (nitroba.pcap) successfully downloaded for analysis with the Info's

a. File Identification

- Filename: Nitroba.pcap
- Format: Wireshark/tcpdump (pcap)
- Encapsulation: ethernet
- Size: 56 MB
- Captured Data: 54 MB
- Capture interfaces: 1
- Capture length: 4096 bytes
- Timestamp precision: Microseconds
- Packets: 94,410

b. Captured Timeline

- First Packet: 22 July 2008, 21:51:07.095278

- Last Packets: 23 July 2008, 02:13:47.046029
- Duration: 15,759.950751 seconds (4h 22 min 40 s)
- Time order: strictly chronological

c. Traffic Characteristics

- Average packet size: 579.07 bytes
- Average packet rate: 5 packets/sec
- Data rate: 3,468 bytes/sec
- Bit rate: 27kbps

Chain of study Custody

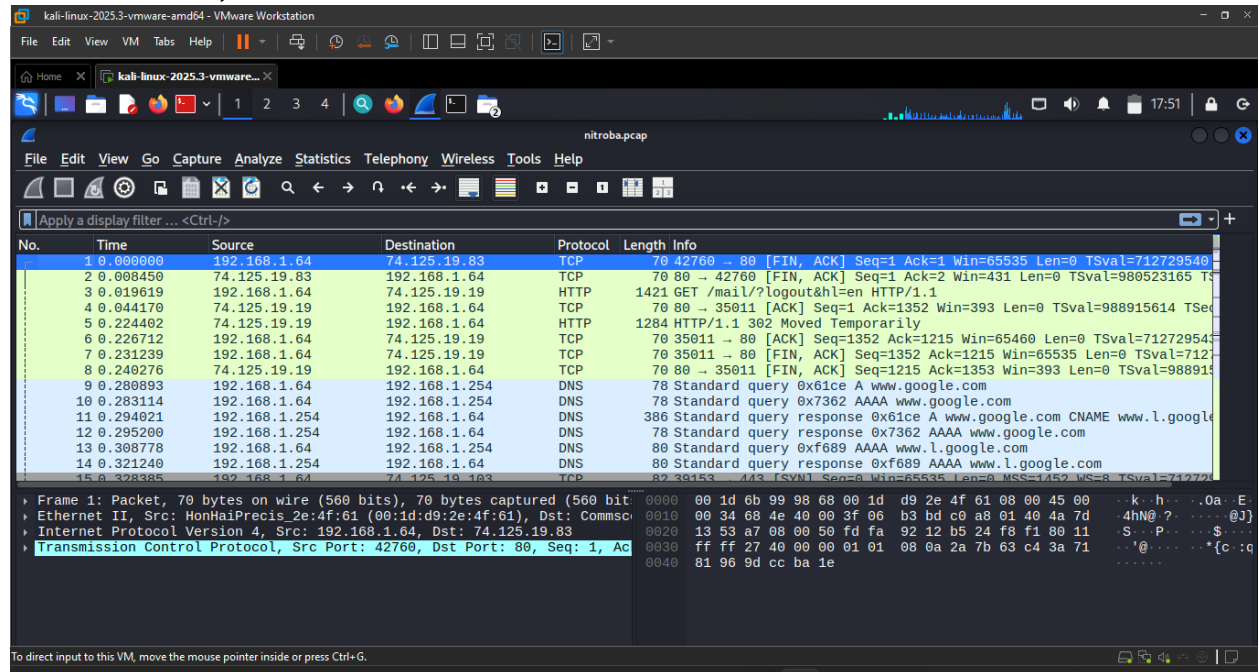
Field	Entry
Case ID	SBT-DF204-CaseStudy-Idriss-Muhamamd-Abdullahi
Examiner	Idriss Muhammad Abdullahi
Date Started	24/09/2026
Evidence File	nitroba.pcap
Source	nitroba State University
Original SHA256	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
Working Copy SHA256	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
Analysis System	Kali Linux VM
Notes	Working copy created before analysis and original copy preserved with no modification

3. Methodology

The investigation was conducted entirely within the Wireshark Graphical user interface. All analysis was performed on a working copy of the original packet capture. Wireshark was used to conduct the analysis and

progressively display filters were applied to identify and examine relevant network communications. Each significant result was validated through packet-level inspections.

3.1 nitro_working.pcap on wireshark showing all protocol (TCP, HTTPs, DNS)



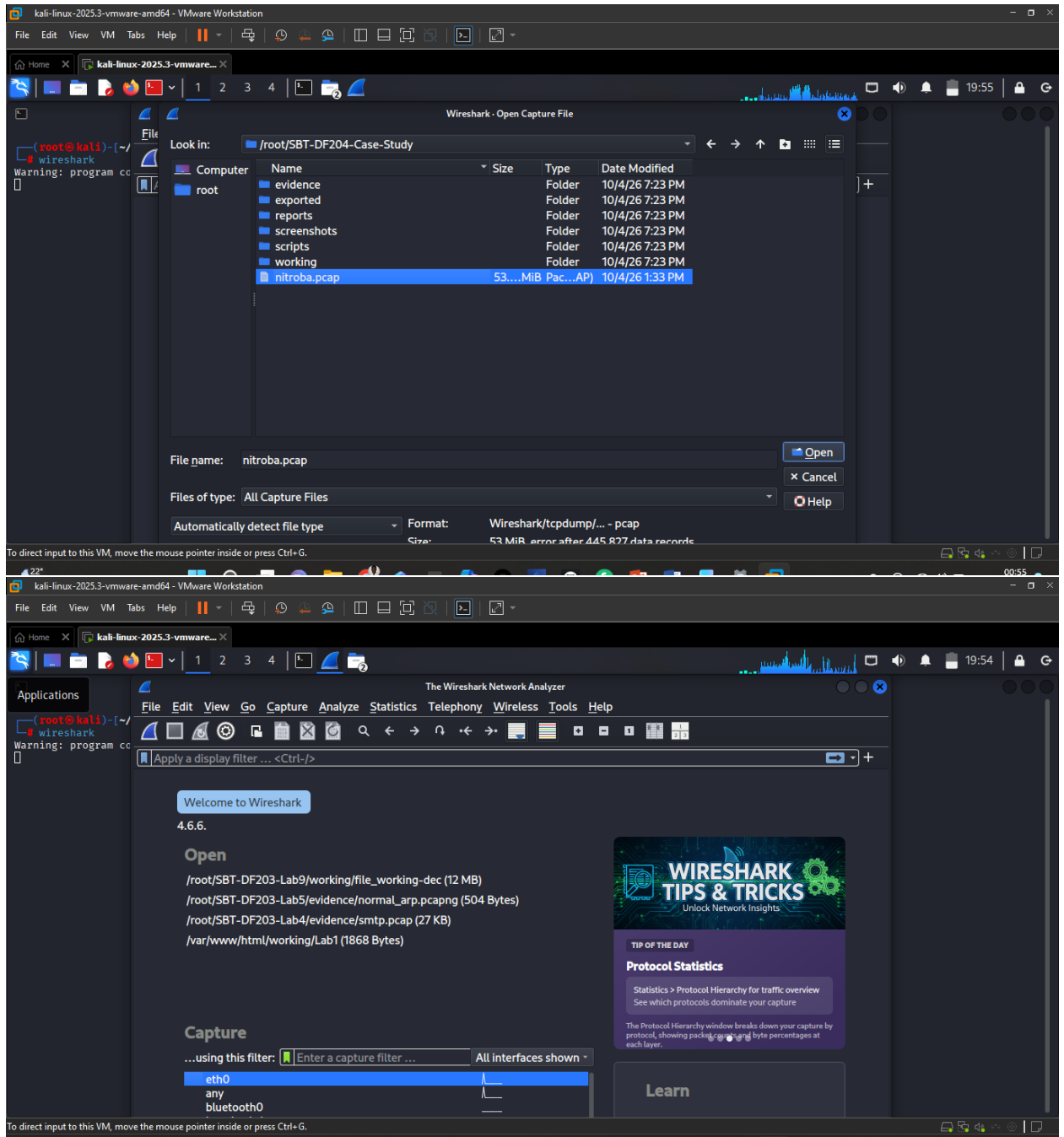
1.2 filtered Sequence

S/N	Display filter	Purposes
1	http.request.method == "GET" && http.host == "www.willselfdestruct.com"	filters traffic to show only HTTP GET requests, which are commonly used when a browser requests a webpage or file. "www.willselfdestruct.com" — limits the results to requests specifically sent to that website.
2	data-text-lines contains "secure anonymous E-mail"	This is for confirming response purpose
3	http.request.method == "POST"	This is used to identify HTTP POST requests in a captured network traffic file
4	ip.src == 192.168.15.4 && ip.dst == 69.25.94.22 &&	This filter is useful for determining whether the client submitted

	http.request.method == "POST"	information to a particular web service.
5	eth.addr == 00:17:f2:e2:c0:ce	This is used to display network packets that involve a specific client's MAC address.
6	eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@"	This is used to search for HTTP packets sent from a specific client device that contain an @ symbol in the HTTP Cookie field.

4. Packet Capture Analysis & Findings

4.1 Opening Wireshark for investigation



4.2 Findings (Investigation Questions)

1. What file did you acquire, and how did you preserve its integrity?

The acquired file was a **PCAP (Packet Capture)** file named **nitroba.pcap**. The file contains captured network traffic collected during the investigation using a packet-capturing tool Wireshark. To preserve

the integrity of the evidence, the original nitroba.pcap file was **saved in its original form without modification**. The cryptographic hash **SHA256** has been generated (990fdc7d78557a3cbf1fe7a1940fec509e3484612771ac3bb9164d76c0265ace).

2. Which Client System contacted the web service?

Filter (http.request.method == "GET" && http.host == "www.willselfdestruct.com"). The first frame (82936) with time 15156.030671 gives out an info (GET /secure/submit HTTP/1.1) request from a source IP of 192.168.15.4 to the destination IP of 69.25.94 with host as www.willselfdestruct.com and multiple GET requests for pages following from the same client request.

The image shows a Wireshark packet capture analysis. The top pane displays a list of captured packets, filtered by 'http.request.method == "GET" && http.host == "www.willselfdestruct.com"'. The list shows several HTTP GET requests from source IP 192.168.15.4 to destination IP 69.25.94.22. The bottom pane shows a detailed view of the first packet (No. 82936, Time 15156.030671), which is an HTTP GET request for /secure/submit. The request headers are visible, including Accept, Referer, Accept-Language, Accept-Encoding, User-Agent, Host, and Connection. The response status is 200 OK, and the server is Apache. The bottom of the window shows the packet details and a search bar.

No.	Time	Source	Destination	Protocol	Length	Info
82936	15156.730593	192.168.15.4	69.25.94.22	HTTP	596	GET /secure/submit HTTP/1.1
82985	15157.030671	192.168.15.4	69.25.94.22	HTTP	355	GET /images/spacer.gif HTTP/1.1
83025	15157.238976	192.168.15.4	69.25.94.22	HTTP	356	GET /images/sm-logo.gif HTTP/1.1
83037	15157.279305	192.168.15.4	69.25.94.22	HTTP	361	GET /images/warning-home.gif HTTP/1.1
83072	15157.435074	192.168.15.4	69.25.94.22	HTTP	356	GET /images/body-bk.gif HTTP/1.1
83087	15157.503801	192.168.15.4	69.25.94.22	HTTP	358	GET /images/btn-send.gif HTTP/1.1
83162	15157.756987	192.168.15.4	69.25.94.22	HTTP	361	GET /images/bridge_small.gif HTTP/1.1
83614	15197.468887	192.168.15.4	69.25.94.22	HTTP	461	GET /secure/success HTTP/1.1
83654	15197.806879	192.168.15.4	69.25.94.22	HTTP	360	GET /images/bk-message.gif HTTP/1.1

Wireshark - Follow TCP Stream (tcp.stream eq 1680) - nitroba.pcap

GET /secure/submit HTTP/1.1
Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, application/x-shockwave-flash, */*
Referer: http://email.about.com/gi/dynamic/offsite.htm?zi=1/XJ&sdn=email&cdn=compute&tm=17&gps=101_1829_788_511&f=00&su=p284.9.336.ip_p504.1.336.ip_&tt=4&bt=1&bts=1&z=1&zu=http%3A/www.willselfdestruct.com/secure/submit
Accept-Language: en-us
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)
Host: www.willselfdestruct.com
Connection: Keep-Alive

HTTP/1.1 200 OK
Date: Tue, 22 Jul 2008 07:24:05 GMT
Server: Apache
Pragma: No-cache
Cache-Control: no-cache
Expires: Thu, 01 Jan 1970 00:00:00 GMT
Set-Cookie: JSESSIONID=0FB2AD5FEEF73F78A1D0A5492813C761; Path=/
Content-Type: text/html; charset=ISO-8859-1
Connection: close
Transfer-Encoding: chunked

1 client pkt, 16 server pkts, 1 turn.

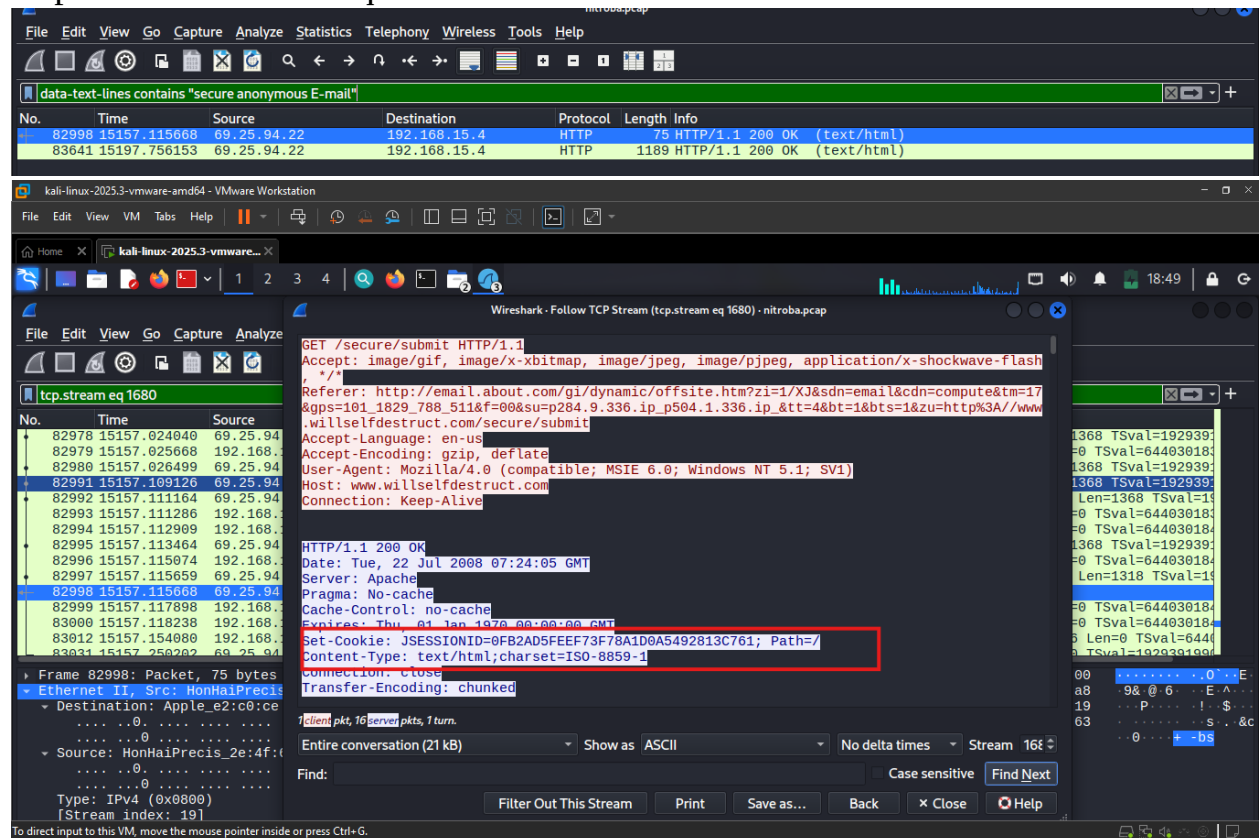
Entire conversation (21 kB) Show as ASCII No delta times Stream 1680

Find: Case sensitive Find Next

Filter Out This Stream Print Save as... Back Close Help

3. What evidence links the client to the harassment message?

Based on the filter sequence, the filter: (data-text-lines contains "secure anonymous E-mail") outputs us only two frames 82998 and 83642 which confirms the response purpose and session cookie JSESSIONID issued in response of the GET requests.



Filter: ip.src == <192.168.15.4> && ip.dst == <69.25.94.22> && http.request.method == "POST". This filters only one frame (83601) with relative time 15197.216422s, with info of POST /secure/submit request url decoded in plain text form:

Form item: to = lilytuckrige@yahoo.com

Form item: from = ""

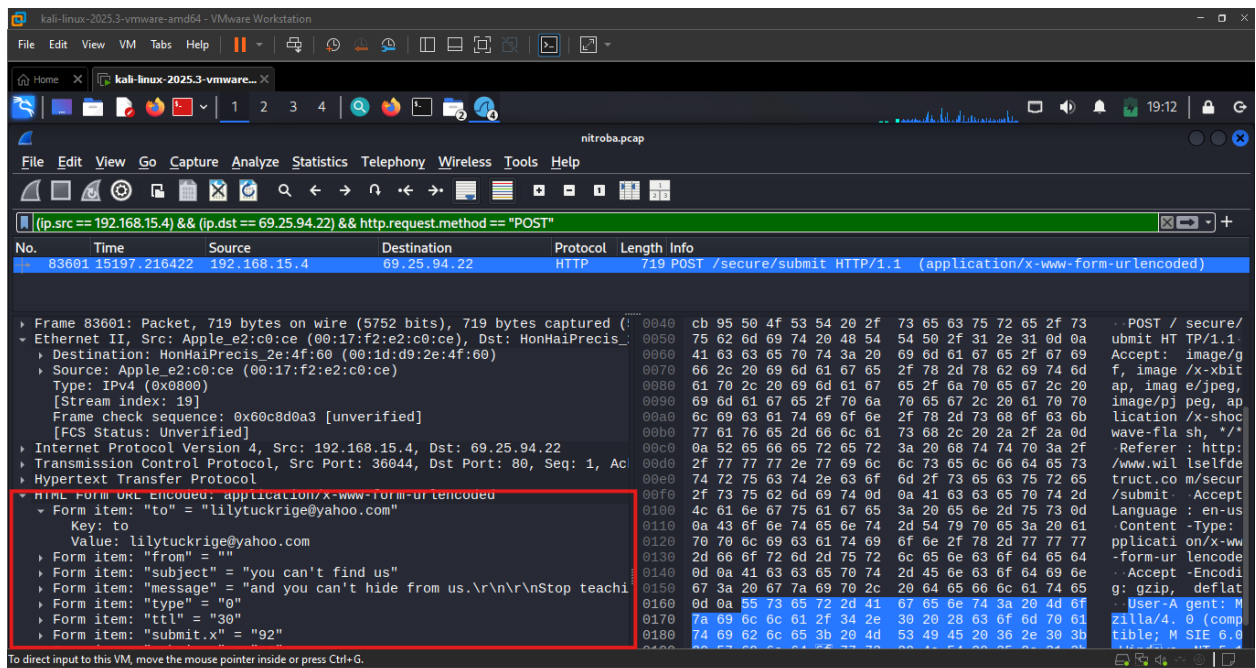
Form item: Subject = "you cant find us"

Form item: message = "and you can't hide from us. \r\n\r\nstop teaching.\r\n\r\nStart running."

Form item: type = "0"

Form item: "ttl" = "30"

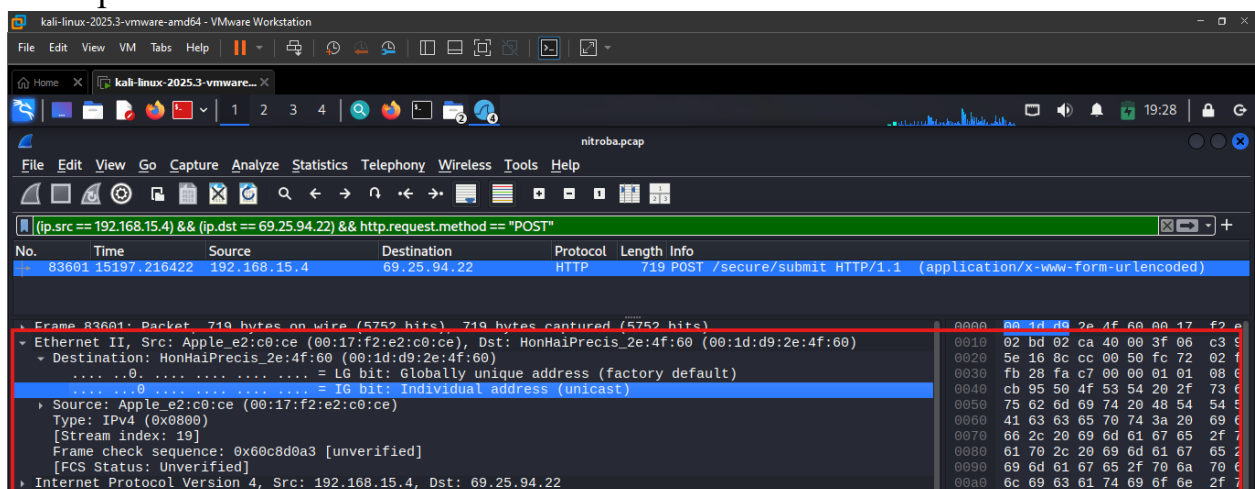
Form item: "submit.x" = "92"



The recipient email address, lilytuckrige@yahoo.com, and the threats content identified in the captured communication are consistent with the complain described in the case scenario. The packet captured provided direct, first hand evidence of the transmission of the message rather than an inference based on circumstantial information

4. Which device made the relevant request?

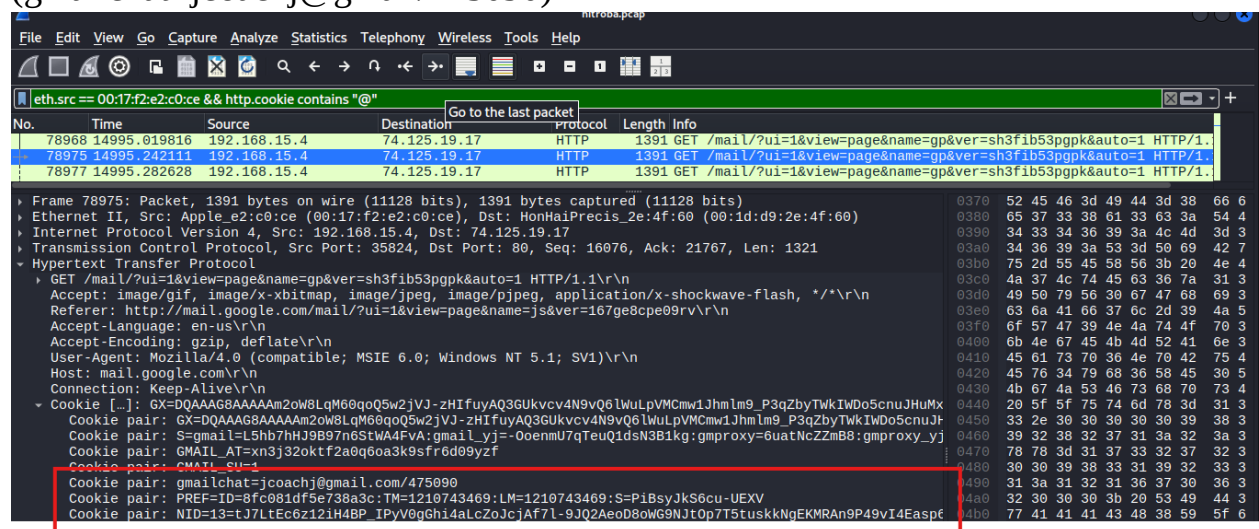
The packet details for frame 83601, with the ethernet layer II expanded, identify the sources as Apple_e2:c0:ce (00:17:f2:e2:c0:ce) and the destination as HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60) which corresponds to the residence's router.



The MAC address 00:17:f2:e2:c0:ce associated with Apple hardware appears as the source ethernet address on the harassment POST itself rather than being inferred from a separate lookup. While the MAC address provides a stronger link to the originating network interface than IP Address alone, it identifies the device not the individual. Because the residence used an open wireless network, the IP address 192,168.15.4 could have been used by another person.

5. Who is associated with the device?

The filter `eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@"` displays multiple packets from a single device to multiple mail services. The frame selected as evidence is frame 78975 with related time 14995.242111s, a request to mail backing a cookie pair (`gmailchat=jcoachj@gmail/475090`)



The cookie found on the same MAC address as the harassment POST contains the google account jcoachj@gmail.com. Its timestamp is approximately 202 seconds before the post, providing a direct packet evidence linking the device to that account.

6. Is that person on the Chem 109 roster?

YES, that person is part of the chem 109 roster by the name Johnny Coach.

7. When did the activity occur?

Wireshark displayed packets timesa s elapsed seconds, while capinfos reported that first packet's absolute timestamp.

Capinfos;

Earliest packet time	2008-07-21 21:51:07.095278
Latest packet time	2008-07-22 02:13:47.046029

8. What conclusion can you defend?

Confidence and Limitations Confidence:

Moderate-to-high confidence that the device and jcoachj@gmail.com account is associated with the harassment message. Confidence is lower when attributing the activity to a specific individual.

Limitations:

The residence used an open wireless network, so another person could have accessed it. MAC addresses can also be spoofed, although no evidence of spoofing was observed. The presence of the Gmail cookie does not prove the same person was operating the device 202 seconds later. Finally, the account has not yet been verified against the Chem 109 roster, so the specific individual associated with it remains unconfirmed.

Evidence Log

Eviden ce ID	Packet/item	Findings	Purpose	imag e
E01	-	Folder structure for packets analysis and preservatio n	To have a controlled working environm ent	Ima ge 1
E02	nitroba.pcap	Packet file (nitroba.pc ap) downloade d	To Analyse and capture evidence	Ima ge 2

E03	Sha256sum_reports/nitroba.pcap.txt	SHA256 of the original packets evidence	Evidence integrity	Image 3
E04	nitroba.pcap	Capinfos with number of packets and working hash matches	Confirm working packets integrity	Image 4
E05	-	All packet list displayed without filter	Launching wireshark for investigation	Image 5

Executive Conclusion

The forensic examination of nitroba.pcap identified network evidence showing that the device using MAC address 00:17:f2:e2:c0:ce and IP address 192.168.15.4 transmitted the threatening message recovered during the investigation. The message was submitted through willselfdestruct.com to lilytuckrige@yahoo.com, with the subject “you can't find us” and the message “and you can't hide from us. Stop teaching. Start running.” These details are consistent with the complaint documented in the case scenario. The same MAC address was also observed shortly before the transmission of the threatening message in traffic containing a Gmail chat cookie identifying jcoachj@gmail.com. This provides an evidential association between the device used to transmit the message and that Google account.

However, the evidence should be interpreted with appropriate limitations. The network was an open residential wireless network, meaning another person could potentially have accessed the network or used the device. Therefore, the findings provide moderate-to-high confidence in linking the

device and the jcoachj@gmail.com account to the network activity, but they do not independently establish which individual was physically operating the device when the message was sent. The association between jcoachj@gmail.com and a specific person could not be independently verified because the Chem 109 roster was not available during this examination. Consequently, roster verification remains an outstanding investigative step before making attribution to a named individual.

Overall assessment:

The packet capture provides strong technical evidence connecting the threatening communication to a specific network device and a specific Google account, while the identification of the individual responsible remains subject to further corroboration.